

LIST OF PAST MIDTERM PROBLEMS

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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1. Akbobek, Bayan, and Uldana meet at a party to which economists and sociologists have been invited. Every attendee at the party is either an economist or a sociologist (but not both). Every economist knows all the other economists, but every sociologist does not know any of the other attendees, whether they are economists or sociologists. The goal of this question is to analyze what do Akbobek, Bayan, and Uldana know about each other's professions.
 - (a) Provide a knowledge space, using only eight states of the world, that describes the situation above. Describe the knowledge space in terms of its equivalent descriptions using:
 - i. Type spaces and signalling functions.
 - ii. Information partitions.
 - (b) What are the atoms of the common knowledge algebra?
 - (c) At what states of the world is the event “There is at least one economist” common knowledge?
 - (d) At what states of the world is the event “There is at least one sociologist” common knowledge?
 - (e) Suppose that Akbobek and Uldana are economists and that Bayan is a sociologist. Show that it is common knowledge between Akbobek and Uldana that the two of them are economists and Bayan is a sociologist, but this fact is not common knowledge among the three of them.
2. Suppose that Mukhamejan (Player 2) owns an oil field in Atyrau. He expects the profit from the oil field to be negative, and therefore intends to sell it. Adiletkhan (Player 1) is of the opinion that the oil field can yield positive profits, and is therefore willing to purchase it (for simplicity, at the price of \$0). Adiletkhan and Mukhamejan arrive at different determinations because they have different information. Suppose the following exchange happens:
 - Adiletkhan: I am interested in purchasing the oil field; are you interested in selling?
 - Mukhamejan: Yes, I am interested in selling; are you interested in purchasing?
 - Adiletkhan: Yes, I am interested in purchasing; are you still interested in selling?
 - And so on, until one of the two parties announces that he has no interest in a deal.

You will model the situation using the following belief space:

$$\begin{aligned} N &= \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} \\ \mathcal{P}_1 &= \{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}, \quad \mathcal{P}_2 = \{\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\}\} \\ \mu(\omega) &= \frac{1}{9}, \quad \forall \omega \in \Omega. \end{aligned}$$

The states in Ω represent all variables affecting the profit of the oil field. Consider the random variable $X : \Omega \rightarrow \mathbb{R}$, given by:

$$X(\omega) = \begin{cases} +1 & \omega \in \{1, 2, 5, 6, 9\}, \\ -1 & \omega \in \{3, 4, 7, 8\}. \end{cases}$$

$X(\omega)$ represents the profit, in billions of tenge, yielded by the oil field at the state of the world ω . At each stage, Adiletkhan will be interested in the deal only if the conditional expectation of X given his information is positive, and Mukhamejan will be interested in the deal only if the conditional expectation of X given his information is negative. The process looks like this:

- Player 1 states whether or not $E[X | \Sigma_1](\omega) > 0$ (implicitly doing so by expressing or not expressing interest in purchasing the oil field). If he says “no” (i.e., his expectation is less than or equal to 0), the process ends here.
- If the process gets to the second stage, player 2 states whether his expectation of X , given the information he has received so far, is negative or not. The information he has includes $P_2(\omega) \in \Sigma_2$, and the affirmative interest of Player 1 in the first stage. If player 2 now says “no” (i.e., his expectation is greater than or equal to 0), the process ends here.

- If the process has not yet ended, player 1 states whether his expectation of X , given the information he has received so far, is positive or not. The information he has includes $P_1(\omega) \in \Sigma_1$, and the affirmative interest of player 2 in the second stage. If player 1 now says “no” (i.e., his expectation is less than or equal to 0), the process ends here.
- And so on. The process ends the first time either Player 1’s expectation of X , given his information, is not positive, or Player 2’s expectation of X , given his information, is not negative.

- (a) How does the process look like at state of the world $\omega = 5$?
- (b) How does the process look like at state of the world $\omega = 1$?
- (c) How does the process look like at state of the world $\omega = 4$?
- (d) How does the process look like at state of the world $\omega = 8$?
- (e) How does the process look like at state of the world $\omega = 9$?

3. Aknur, Rysty, and Yerbol are standing first, second, and third in a line, respectively. Each one sees the persons standing in front of them. Prof. Alejandro announces: “I have three red hats and two white hats. I will place a hat on the head of each one of you.” After Prof. Alejandro places the hats, he asks Yerbol (who can see the hats worn by Rysty and Aknur) if he knows the color of the hat on his own head. Yerbol replies “no”. He then asks Rysty (who sees only Aknur’s hat) whether she knows the color of the hat on her own head, and she also replies “no”. Finally, he asks Aknur (who cannot see any of the hats) if she knows the color of the hat on her own head.

- 0.a Construct a set of states Ω that contains 7 states of the world and describes the situation above.
- 0.b What are the partitions of Aknur, Rysty, and Yerbol after Prof. Alejandro’s announcement and before he asked Yerbol whether he knows the color of his hat?
- 0.c What are the partitions of Aknur and Rysty after Yerbol’s response and before Rysty responded to Prof. Alejandro’s question?
- 0.d What is Aknur’s partition after hearing Rysty’s response?
- 0.e What is Aknur’s answer to Prof. Alejandro’s question? Does this answer depend on the state of the world, that is, on the colors of the hats that the three wear?

4. Professor Alejandro has disappeared without a trace! The students of ECON101 and ECON403 are two of the prime suspects in this academic whodunit. Why, you ask? Well, Professor Alejandro dared to give an ECON101 exam so tough that it left a trail of bruised egos and tears longer than an infinitely elastic demand curve! And as for ECON403, they had a midterm looming on the horizon. Witnesses also claim that Yernur, a past student from Prof. Alejandro, was caught yelling at him, “I’m furious! I was one of your best students, but my name never got to be the hero in one of your problem-solving adventures!”

Detective Nurdaulet and Detective Nurassyl are assigned to the case. They were trained at the same police academy. Their instruction consists of a well-defined rule specifying who to arrest given the clues that they have discovered. The two detectives are ordered to conduct an independent investigation. They promise not to share any data gathered from their research, and begin their sleuthing in different corners of NU. Suddenly the detectives are asked to appear and announce who they plan to arrest. Neither has had the time to complete a full investigation, so they each have gathered different clues. They meet on their way to the police station. Recalling their pledges, they do not tell each other a single discovery, or even a single reason why they were led to their respective conclusions. But they do tell each other who they plan to arrest. Hearing the other’s opinion, each detective may change his mind and give another opinion.

Suppose that the above situation and all of its specifics can be modeled with the following Belief space:

$$N = \{1, 2\}, \quad \Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathcal{P}_1 = \{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}, \quad \mathcal{P}_2 = \{\{1, 2, 3, 4\}, \{5, 6, 7, 8\}, \{9\}\}$$

$$\mu(\omega) = \frac{1}{9}, \quad \forall \omega \in \Omega.$$

Each $\omega \in \Omega$ represents all facets of the investigation related to the kidnapping. Detective Nurdaulet is player 1 and detective Nurassyl is player 2. Their information partitions represent the outcomes of their individual investigations. Consider the random variable $X : \Omega \rightarrow \mathbb{R}$:

ω	1	2	3	4	5	6	7	8	9
$X(\omega)$	17	-7	-7	0	17	-7	-7	-7	0

The different values of this random variable represent who is determined to be guilty at each state of the world: 17 for the ECON101 students, -7 for the ECON403 students and 0 for Yernur.

In addition to their training at the police academy, both Detective Nurdaulet and Detective Nurassyl are extremely smart and logical, they both know this, and they also took ECON403 with Prof. Alejandro. Suppose that Detective Nurdaulet and Detective Nurassyl talk long enough. Show that we can be sure that both detectives will announce the same suspect(s) at the station.

5. In our galaxy, far beyond the reaches of Earth, lies the peculiar planet of **NUpiter**, home to the highly logical and tradition-bound **NUpiterians**. NUpiter is inhabited by K NUpiterians. These beings pride themselves on recognizing intellect and work ethic in others, yet, due to an ancient and unshakable belief system, *no NUpiterian may ever know whether they themselves are smart and hardworking!* Can you believe it?

Each NUpiterian can easily recognize intelligence and hard work in others, but their strange culture dictates that no NUpiterian can recognize intelligence and hard work within themselves. If a NUpiterian ever realizes that they are smart and hardworking, their religion compels them to perform a very scary ritual at noon on the following day: they must ascend to another plane of existence—a realm so terrifying that it is only spoken of in hushed whispers: “*Real Life*.”

Once in *Real Life*, a NUpiterian faces horrors beyond imagination: responsibilities, decision-making, and worst of all, getting a job. For these reasons, NUpiterian tradition dictates that they must live in carefully maintained ignorance, ensuring that no one ever meets this fate.

All NUpiterians are highly logical and devout in their customs, and they all know that each other is also highly logical and devout (and they all know that they all know that each other is highly logical and devout, and so forth).

Among the K NUpiterians, it turns out that N of them are smart and hardworking, where $N \leq K$, and the remaining $K - N$ are not. However, the NUpiterians are not initially aware of this distribution (each of them, of course, recognizes the intelligence and work ethic of the other $K - 1$ NUpiterians, but they do not know their own status).

One fateful day, a *dashing and daring explorer*, Prof. Alejandro, arrives on NUpiter. A fearless traveler with an *unparalleled knack for chaos*, he is instantly recognized as smart and hardworking by the locals, so he quickly gains their trust and is invited to deliver an address to the entire civilization.

Aware of their cultural taboos, but always with a taste for chaos, Prof. Alejandro mischievously remarks:

Ah, it’s wonderful to come in contact with other smart and hardworking beings such as myself! I was starting to think I might be the only one in this region of the galaxy!

The moment these words leave his mouth, a horrified silence falls over the crowd. The NUpiterians, bound by logic, begin to think.

What effect, if anything, does this chaotic remark by Prof. Alejandro have on the NUpiterians?

- (a) Suppose first that $N = K = 3$. Let S denote “hardworking and studious”, and $\neg S$ denote “Not hardworking or studious”. Using the previous notation, represent this situation as a knowledge space with $2^3 = 8$ states of the world.
 - (b) Use the knowledge space framework to prove that after three days, all three NUpiterians ascend to *Real Life*.
 - (c) What new information did Prof. Alejandro introduce by making his announcement?
 - (d) Generalize this result for arbitrary values of N and K where $N \leq K$.
6. Suppose that there are two players, 1, and 2. The set of states is $\Omega = \{1, 2, \dots, 12\}$ and the common prior of the two players is $\mu(\omega)$. Assume $\mu(\omega) > 0$ for all $\omega \in \Omega$.

Each player i observes, privately, the realization of the random variable $X_i : \Omega \rightarrow \mathbb{R}$, where these random variables are defined as follows:

$$X_1(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2, 3, 4\} \\ 2 & \text{if } \omega \in \{5, 6, 7, 8\} \\ 3 & \text{if } \omega \in \{9, 10, 11, 12\} \end{cases} \quad X_2(\omega) = \begin{cases} 1 & \text{if } \omega \in \{1, 2\} \\ 2 & \text{if } \omega \in \{3, 4\} \\ 3 & \text{if } \omega \in \{5, 6\} \\ 4 & \text{if } \omega \in \{7, 8\} \\ 5 & \text{if } \omega \in \{9, 10\} \\ 6 & \text{if } \omega \in \{11, 12\} \end{cases}$$

Let $A \in 2^\Omega$ be an event. Define $Q_i^0(\omega) = E[\mathbb{1}_A | X_i](\omega)$ for every $\omega \in \Omega$, where $\mathbb{1}_A$ is the indicator function for event A . Then, for $t = 1, 2, \dots$, define, for every $\omega \in \Omega$:

$$Q_1^t(\omega) = E[\mathbb{1}_A | X_1, Q_2^0, \dots, Q_2^{t-1}](\omega), \quad Q_2^t(\omega) = E[\mathbb{1}_A | X_2, Q_1^0, \dots, Q_1^{t-1}](\omega).$$

Answer the following True/False questions. For the answer to be valid, you should provide an argument/proof. Hint: All proofs here are very short.

- (a) True or False. For any μ and any event A , $\lim_{t \rightarrow \infty} Q_i^t$ exists for $i = 1, 2$ and the limits are equal.
- (b) True or False. For any μ and any event A , the players always end up with the same information partitions at the end of the process.

7. Consider the following Arabian proverb:

There are four sorts of men:

He who knows not and knows not he knows not: he is a fool—shun him;

He who knows not and knows he knows not: he is simple—teach him;

He who knows and knows not he knows: he is asleep—wake him;

He who knows and knows he knows: he is wise—follow him.

- (a) Let Ω be a finite set of states of the world. Let $N = \{F, S, A, W\}$ denote the set of agents, where F denotes the fool, S the simple, A the asleep, and W wise. Suppose that for each agent, we have a knowledge operator $L_i : 2^\Omega \rightarrow 2^\Omega$ that follows the setup of the proverb. For each agent $i \in N$, which one of Kripke's axioms to they violate?
- (b) Can any of the knowledge operators from the each of the previous agents come from a standard model with information partitions?
- (c) Define a *possibility correspondence* as a function $P : \Omega \rightarrow 2^\Omega$ that is not necessarily partitional. Define its induced knowledge operator as usual

$$KA = \{\omega \in \Omega : P(\omega) \subseteq A\}.$$

Consider $\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$ and the following possibility correspondence:

$$P(\omega_1) = \{\omega_1, \omega_2\}$$

$$P(\omega_2) = \{\omega_2, \omega_3\}$$

$$P(\omega_3) = \{\omega_3\}$$

$$P(\omega_4) = \{\omega_4\}$$

Consider an agent whose knowledge operator K is induced by the previous possibility correspondence. Can this agent be the fool? Can this agent be the simple? Can this agent be asleep? Can this agent be wise?